Variables, types, and expressions

Module 1 - Lesson 2

Overview

Variables name values in your program. Python is dynamically typed, so any name can hold any type, but knowing the core types keeps your data predictable.

Key Concepts

- Assign values with = ; names are case-sensitive and conventionally use snake_case.
- Core types: int, float, str, bool, list, dict, tuple, set, and None.
- type() reveals the type of any value; use isinstance() for checks.
- Strings support indexing, slicing, f-strings, and dozens of methods.
- Expressions combine values with operators; order follows standard precedence.
- Immutability matters: strings and tuples are immutable, lists and dicts are mutable.

Example

```
price = 19.99
count = 3
label = f"Total: {price * count:.2f}"
print(label)  # Total: 59.97

items = ["a", "b", "c"]
print(items[1:])  # [’b’, ’c’]
```

Summary

Variables name values; types describe what a value can do. Strings, numbers, and containers cover most data, and f-strings make formatting readable.

Practice Checklist

- Write expressions that combine strings and numbers and format the output.
- Slice a list and a string and predict the results.
- Use type() to inspect the types of a few values you create.